

MALAV PATEL

1307 Kingsbury Dr Unit C, Hanover Park, IL 60133 | (630) 359-1224 | mpatel636@gatech.edu | <https://malav-p.github.io/>

Analytical and detail-oriented student passionate about industry trends in machine learning and scalable ML systems. Experience with projects using transformers, VAEs, audio codecs, LSTMs, diffusion models, and CUDA programming. Eager to find work applying coursework in statistics, linear algebra, optimization, and machine learning to industry problems.

EDUCATION

Master of Science, Computer Science | Georgia Institute of Technology | GPA 4.0

January 2024 - April 2026

Relevant coursework: Statistical ML, Mathematical Foundations of ML, Computational Data Analysis, Optimization

Doctor of Philosophy, Aerospace Engineering | Georgia Institute of Technology | GPA 3.88

August 2022 - April 2026

Optimization methods for constellation design in cislunar space

EXPERIENCE

Graduate Research Assistant | Space Systems Optimization Group | Atlanta, GA

August 2022 – Present

Utilizing/developing novel optimization schemes for satellite constellation design in cislunar space.

- Cislunar Satellite Constellation Design via Integer Linear Programming - [Paper](#)
- Concurrent Optimization of Satellite Phasing and Tasking for Cislunar Space Situational Awareness – [Paper](#)

Machine Learning Engineer | On Device GPT Inference | Atlanta, GA

December 2023 – August 2024

Built OpenAI GPT2 from scratch in C++. Optimized memory management via in-place memory transformations, KV-caching, inference optimizations, reducing inference latency by 16x. Strong understanding of transformer model architecture. [Code](#)

Machine Learning Engineer | CNN Training and Inference | Atlanta, GA

December 2021 – June 2022

Designed a library from scratch in C++ for the implementation of a convolutional neural network. Implemented trained model on MNIST dataset to 97.58% accuracy. Accelerated computations with BLAS routines. Accelerated training 3x via GPU programming using CUDA. Interfaced library with OpenCV to predict digits shown on a live video feed. [Code](#)

Machine Learning Engineer | Patent Classification | Atlanta, GA

January 2025 – April 2025

Achieved 82% precision on the patent classification task using ModernBERT, matching existing state-of-the-art performance with 2x the training throughput.

Machine Learning Engineer | Audio Modeling | Atlanta, GA

May 2025–June 2025

Established an open-source implementation of SoundStream, a neural audio codec, within two weeks. Optimized training times via hyperparameter tuning and released trained checkpoints.

SKILLS

PyTorch, Tensorflow, C++, CUDA

Experience with: Neural Audio Codecs (SoundStream, VQ-VAE), Sequence Classification (BERT / modernBERT), Language Generation GPT, Mixture of Experts), Image Generation (VAE, diffusion, LDM)